AI-POWERED ASSISTIVE SYSTEM FOR VISUALLY IMPAIRED PEOPLE

A PROJECT REPORT

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ABSTRACT

An AI-powered assistive system for visually impaired individuals that integrates computer vision, natural language processing, and speech synthesis into a unified framework to provide real-time assistance across multiple everyday tasks including obstacle detection, currency recognition, text reading, social interaction, and environmental understanding.

Visual impairment affects more than 285 million people worldwide, of whom nearly 39 million are completely blind. Existing assistive technologies are often limited to single-functional solutions and do not address region-specific requirements such as Indian currency recognition. To overcome these limitations, the proposed system brings together multiple AI technologies — YOLOv5-based obstacle detection, Darknet YOLOv4-based Indian currency recognition, DeepFace-based facial and emotion recognition, EasyOCR-based text reading, and BLIP-based scene captioning — under a single integrated platform.

The system's conversational interaction is powered by a Dialogflow chatbot with Transformer models, and all outputs are made accessible through eSpeakNG text-to-speech synthesis. A Flask-based API manages communication between modules, and MongoDB handles user-specific data storage. Experimental observations indicate enhanced usability with the system achieving 94–98% accuracy across core modules, demonstrating its potential to significantly improve the independence and quality of life for visually impaired users.

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LIST OF ABBREVIATIONS

ABBREVIATION	FULL FORM
AI	Artificial Intelligence
API	Application Programming Interface
BLIP	Bootstrapping Language-Image Pre-training
CER	Character Error Rate
CNN	Convolutional Neural Network
CTC	Connectionist Temporal Classification
DNN	Deep Neural Network
EasyOCR	Easy Optical Character Recognition
eSpeakNG	Extended Speech Application Next Generation
IoU	Intersection over Union
LSTM	Long Short-Term Memory
ML	Machine Learning
NLP	Natural Language Processing
NMS	Non-Maximum Suppression
OCR	Optical Character Recognition
PANet	Path Aggregation Network
REST	Representational State Transfer
SPP	Spatial Pyramid Pooling
TTS	Text-to-Speech
VGG	Visual Geometry Group
ViT	Vision Transformer
YOLO	You Only Look Once

CHAPTER 1
INTRODUCTION

1.1 DOMAIN OVERVIEW

Visual impairment is a major public health challenge that affects millions of individuals across the world. According to the World Health Organization, more than 285 million people are living with some form of visual impairment globally, of whom nearly 39 million live in complete blindness [1]. The impact of visual impairment extends well beyond the inability to see — it significantly restricts a person's capacity for independent living, social participation, communication, and safe mobility. Everyday tasks that sighted individuals perform effortlessly, such as reading a medicine label, identifying a currency note, recognising a familiar face in a crowd, or crossing a busy road, can be profoundly challenging for those with visual impairment.

Assistive technologies have long played an important role in supporting people with visual impairments. Traditional aids such as white canes, Braille systems, and screen readers provide limited support [2]. However, these solutions are largely limited in scope and do not address the full range of daily-life challenges faced by visually impaired users. Furthermore, multilingual accessibility and inclusive assistive environments have been explored in recent studies, highlighting the importance of speech-based feedback systems for diverse users [3].

The rapid advancement of Artificial Intelligence, computer vision, natural language processing, and speech synthesis technologies over the past decade has created a significant opportunity to develop next-generation assistive systems. Deep learning models for object detection, facial recognition, optical character recognition, and vision-language understanding have achieved human-level or near-human-level performance on many perceptual tasks. When integrated thoughtfully, these capabilities can form the basis of a powerful, unified assistive system that addresses a wide spectrum of challenges faced by visually impaired individuals in their daily lives.

1.2 PROJECT OVERVIEW

The AI-Powered Assistive System for Visually Impaired People is a unified intelligent framework designed to help visually impaired individuals perceive, interpret, and interact with their surrounding environment in real time. The system integrates seven core functional modules — obstacle and object detection, Indian currency recognition, facial and emotion recognition, printed and handwritten text reading, scene understanding, speech-based feedback, and conversational chatbot support — into a single cohesive platform.

Obstacle and object detection is implemented using YOLOv5 and Darknet YOLO-based models that process live camera frames and alert the user about nearby objects through voice feedback. Indian currency recognition is handled by a Darknet YOLOv4 model trained specifically on a dataset of Indian banknote images covering all major denominations from Rs. 10 to Rs. 2000. Facial recognition and emotion detection are achieved using DeepFace in combination with OpenCV, enabling the system to identify registered individuals and classify their emotional state in real time.

Text reading is performed by EasyOCR, which employs a CNN and LSTM-based architecture to support recognition of both printed and handwritten text. Environmental scene understanding is delivered through BLIP (Bootstrapping Language-Image Pre-training), a transformer-based vision-language model that generates natural-language captions describing the user's surroundings. All outputs are converted to accessible audio using eSpeakNG, a compact offline text-to-speech engine with approximately 200 ms latency. A Dialogflow-based chatbot with Transformer models provides conversational interaction, and a Flask REST API with MongoDB backend handles module integration and persistent data storage.

A classification layer first determines the nature of the input: if an Indian currency note is detected, the currency recognition pipeline is activated; if a human face is identified, the facial and emotion recognition pipeline is invoked; otherwise, scene captioning is performed. This selective activation ensures efficient use of computational resources, making the system suitable for deployment on mobile and embedded platforms.

1.3 CHALLENGES IN THE DOMAIN

Developing an integrated AI-powered assistive system for visually impaired users presents several significant technical and practical challenges:

- **Multi-Module Integration:** Combining multiple deep learning models — including YOLO-based detection, DeepFace-based recognition, transformer-based captioning, and OCR — into a single coherent real-time pipeline requires careful orchestration of computational resources. Each module has different latency and memory requirements, making unified integration a non-trivial engineering challenge.
- **Indian Currency Recognition:** Most existing currency recognition systems are designed for internationally circulated banknotes and do not support Indian Rupee denominations. The diversity of Indian currency note designs, combined with varying illumination conditions and backgrounds encountered in real use, makes this a specialised and underserved recognition problem.
- **Real-Time Performance on Resource-Constrained Devices:** The system is intended to run on mobile devices or embedded platforms with limited computational resources. Ensuring real-time inference speed while maintaining high recognition accuracy across multiple modules is a fundamental design challenge.
- **Accuracy under Real-World Conditions:** Facial recognition and emotion detection must operate reliably under uncontrolled real-world conditions including varying lighting levels, different head poses, partial occlusions, and diverse facial expressions.
- **Offline Usability:** In many parts of India, consistent internet connectivity cannot be assumed. The system must include offline-capable components to ensure usability in low-connectivity environments.
- **Region-Specific Language and Speech:** India is a linguistically diverse country. The speech output and text recognition capabilities must support multiple Indian regional language.

1.4 MOTIVATION

Visual impairment does not merely affect what a person can see — it profoundly shapes how a person experiences the world, maintains relationships, and participates in society. Everyday activities that most people perform without conscious effort, such as reading printed text on a product label, identifying the denomination of a currency note before paying at a shop, recognising a friend or family member who approaches them, or understanding what kind of environment they have entered, become sources of uncertainty and dependency for visually impaired individuals.

The central motivation of this project is to address these daily-life challenges through a unified, intelligent, and voice-driven assistive system. Existing solutions are predominantly single-functional: a navigation aid does not help with currency identification; a currency recognition app does not provide scene descriptions; a screen reader does not recognise faces. This fragmented landscape means that a visually impaired user must juggle multiple separate tools, each with its own interface, learning curve, and limitations.

A particular gap that motivates this project is the absence of Indian currency recognition in available assistive technology solutions. Financial independence — the ability to independently identify banknotes and complete everyday transactions without assistance — is a critical aspect of daily autonomy that has been systematically overlooked by globally developed assistive tools. This project directly addresses that gap by developing and integrating a dedicated Indian currency recognition module trained on real Indian banknote data.

1.5 CONTRIBUTION OF THE PROJECT

The key contributions of this project are:

- Development of a multi-module AI-powered assistive system integrating seven distinct functional capabilities — obstacle detection, Indian currency recognition, facial recognition, emotion detection, text reading, scene captioning, and conversational interaction — within a single unified framework.

- Implementation of a specialised Indian currency recognition module using Darknet YOLOv4 trained on a custom dataset of 2,050 Indian banknote images covering all major denominations, achieving 98.1% recognition accuracy.

- Integration of facial identification and emotion recognition using DeepFace and OpenCV for real-time social awareness, achieving a 96.8% F1-score across six emotion classes.

- Deployment of BLIP transformer-based scene captioning to provide contextually rich natural-language descriptions of the user's surrounding environment.

- Implementation of offline-capable eSpeakNG text-to-speech synthesis with approximately 200 ms output latency, ensuring real-time accessible voice feedback without internet dependency.

- Design of a modular Flask REST API with MongoDB backend enabling scalable integration of all system components with persistent user data storage.

1.6 ORGANIZATION OF THE REPORT

This report is structured into nine chapters, each covering a specific aspect of the AI-Powered Assistive System.

Chapter 1 introduces the domain overview, project overview, key challenges, motivation, contributions, and overall organization of the work. Chapter 2 presents a comprehensive literature review of existing assistive technologies and related AI-based approaches. Chapter 3 focuses on system analysis, including the problem definition, proposed solution, software components, and use cases. Chapter 4 explains the system design, detailing the overall architecture, individual modules, and system workflow supported by sequence and activity diagrams. Chapter 5 describes the implementation process, including development tools, algorithms used, step-by-step implementation, and experimental setup. Chapter 6 discusses the system testing approach, evaluation metrics, and test cases used to validate the system. Chapter 7 showcases the outputs of each module with relevant visual examples and explanations. Chapter 8 provides an analysis of the results and compares the system's performance with existing solutions. Finally, Chapter 9 concludes the project and outlines possible directions for future enhancements.

CHAPTER 2
RELATED WORK

2.1 LITERATURE REVIEW

The use of artificial intelligence in assistive technology for visually impaired people has attracted considerable research attention over the past decade. Early work was primarily focused on navigation assistance and obstacle avoidance using computer vision techniques. Object detection-based systems employing deep learning algorithms such as YOLO and SSD demonstrated promising results in identifying obstacles and providing real-time auditory feedback to users.

Joshi [4] proposed an AI-based navigation system that enhanced spatial awareness by detecting obstacles and warning the user through voice prompts. The system used YOLO-based object detection on live camera frames and classified objects by distance to determine safe and unsafe zones. While the system achieved reliable obstacle alerts, it was limited exclusively to navigation tasks and did not address any other assistive needs such as text reading, currency identification, or social interaction.

Tai [5] developed an Android-based application capable of detecting vehicles and estimating their distance from the user, aimed at improving the safety of outdoor navigation for visually impaired individuals. The system used deep learning-based vehicle detection and a distance estimation heuristic based on bounding box dimensions. Although effective for road safety, the system operated as a single-purpose tool and was not extendable to other daily-life assistive functions.

Research subsequently expanded beyond navigation to include text recognition and Indian currency identification. Wearable and mobile assistive systems began combining object detection with optical character recognition to help users interact with printed content and financial instruments.

Joshi [6] presented DRISHTI, a wearable visual assistant that combined object detection and OCR methods to identify printed text and currency denominations, allowing users to complete daily tasks more independently. The system demonstrated the practical value of combining multiple capabilities within a wearable form factor, though it remained limited in terms of scene understanding and social interaction features.

Jaiswal and Singh [7] proposed an Indian currency recognition framework specifically designed for visually impaired users. The system used deep learning models including CNN and CNN-LSTM architectures to identify Indian banknote denominations under different illumination conditions and backgrounds, achieving an accuracy of 96.2%. This work directly addressed the underserved problem of Indian Rupee recognition but was developed as an isolated module without integration with navigation, text reading, or speech interaction capabilities.

More recent research has shifted focus towards multimodal interaction, scene understanding, and conversational assistance to deliver richer user experiences. Transformer-based vision-language models have introduced the ability to generate semantically rich natural-language descriptions of visual scenes, which is a significant advancement for environmental awareness assistance.

Li [8] introduced BLIP (Bootstrapping Language-Image Pre-training), a transformer-based vision-language model capable of generating contextually accurate captions for images. BLIP was shown to capture complex visual-semantic relationships and produce natural-language descriptions suitable for scene understanding applications. Its pre-training on large-scale datasets enables strong generalisation across diverse scene types, making it well-suited for environmental description in assistive systems.

Jain [9] explored speech-based conversational interfaces for assistive technologies, demonstrating that NLP-driven chatbot integration can provide visually impaired users with a natural and intuitive means of interacting with assistive systems. Their work highlighted the importance of intent recognition, contextual response generation, and low-latency speech output in building effective conversational assistants.

Subramanian [10] investigated Edge-AI architectures for inclusive assistive systems, proposing methods to reduce computational dependency on cloud servers. By deploying lightweight AI models on edge devices, their approach significantly reduced inference latency and enabled reliable operation in low-connectivity environments, which is an important practical consideration for assistive deployment in rural or network-limited areas of India.

Cloudin and Manikandan [11] developed an intelligent vision assistance framework using deep learning techniques for object identification and spatial awareness. Cloudin [12] further proposed a haptic-enabled Braille communication system for visually impaired users that integrated physical feedback with digital text conversion. These studies demonstrate the growing diversity of AI approaches being applied to assistive technology, though each continues to address only a limited subset of daily-life challenges.

2.2 INFERENCE OF LITERATURE REVIEW

The literature review shows that assistive technology for visually impaired individuals has progressively advanced from single-purpose navigation tools to more capable multi-function systems. However, the scope of integration remains limited in existing approaches, with most systems combining at most two or three capabilities. Table 2.1 summarises the key contributions from the reviewed literature.

Table 2.1 Inference of Literature Review

Ref.	Work	Contribution	Limitation
[4]	AI Navigation System	Obstacle detection with voice alerts	Navigation only
[5]	Guardian Angel	Vehicle detection for road safety	Single-purpose tool
[6]	DRISHTI Wearable	Text and currency reading in wearable	No emotion or chat support
[7]	Indian Currency Framework	Indian banknote ID at 96.2% accuracy	Isolated; no speech integration
[8]	BLIP Vision-Language	Rich natural-language scene captions	Not a complete assistive system
[9]	Speech Interfaces for AT	Natural voice interaction for assistive	No vision modules
[10]	Edge-AI Assistive	Offline-capable inference	No multi-module integration
[11]	Intelligent Vision	Object ID and spatial awareness	No currency or social modules
[12]	Braille Haptic	Braille feedback with text recognition	Text only; no scene or social

2.3 EXTRACTION FROM LITERATURE REVIEW

From the literature review, several important observations can be drawn that directly inform the design of the proposed AI-Powered Assistive System.

First, YOLO-based object detection models — particularly YOLOv5 and Darknet YOLOv4 — have demonstrated strong real-time performance for obstacle detection and currency classification respectively. The use of YOLOv4 specifically for Indian currency recognition, as demonstrated by Jaiswal and Singh, confirms the suitability of this architecture for denomination-level classification of Indian banknotes.

Second, the integration of DeepFace for facial recognition and emotion detection offers a practical and accurate solution for social awareness, supporting multiple backend models and providing both face verification and facial expression recognition in a single library.

Third, BLIP's transformer architecture provides contextually accurate scene descriptions that go significantly beyond earlier CNN-based captioning models. Its open-source availability makes it practical for integration without requiring custom pre-training on large-scale datasets.

Fourth, EasyOCR provides robust support for both printed and handwritten text across multiple scripts, which is particularly important in the Indian context where documents may contain text in multiple regional languages.

Fifth, the literature consistently highlights offline capability as critical for assistive systems designed for India. The selection of eSpeakNG for text-to-speech synthesis, which operates entirely offline, directly addresses this requirement and distinguishes the proposed system from solutions depending on cloud-based TTS APIs.

CHAPTER 3
SYSTEM ANALYSIS

3.1 PROBLEM DEFINITION

The primary problem addressed by this project is the absence of a comprehensive, region-adaptive assistive system for visually impaired individuals that integrates multiple AI-driven capabilities into a single unified, offline-capable platform. Existing assistive solutions suffer from the following well-documented limitations:

- **Fragmentation**: Current solutions are predominantly single-functional. A visually impaired user wishing to navigate safely, identify a currency note, recognise a friend, read a label, and understand an unfamiliar environment must use four or five different applications, each with its own interface, activation method, and learning curve.
- **Absence of Indian Currency Recognition:** None of the widely available assistive applications supports recognition of Indian Rupee banknotes, creating a critical gap in financial independence for visually impaired individuals in India.
- **Cloud Dependency:** Many existing AI-based assistive applications rely on cloud-based computation for inference. In India, where rural and semi-urban areas frequently have unreliable internet connectivity, this dependency severely limits usability.
- **Lack of Social Awareness:** No existing single-platform solution integrates both facial identification and emotional state detection, which are fundamental to enabling meaningful social interaction for visually impaired users.
- **Limited Regional Language Support**: Existing OCR and TTS solutions often prioritise English and major global languages, leaving users who communicate primarily in Indian regional languages underserved.

3.2 PROPOSED SOLUTION

To address the identified limitations, this project proposes an AI-Powered Assistive System that integrates seven core modules within a unified, voice-driven, offline-capable framework [13]. The proposed solution is characterised by the following design principles:

- **Unified Multi-Modal Architecture:** All seven functional modules are accessible through a single interface. Users interact with the system entirely through voice commands and receive all information as spoken audio output.

- **Hierarchical Decision Framework:** A classification layer at the input stage determines the nature of the visual input and routes it to the appropriate processing module, ensuring computational efficiency.
- **Offline-First Design:** Core modules including obstacle detection, currency recognition, text reading, and speech synthesis operate entirely offline, with no dependency on internet connectivity.
- **Region-Adaptive Indian Currency Recognition:** A dedicated currency recognition module, trained specifically on Indian Rupee banknotes spanning all major denominations from Rs. 10 to Rs. 2000, provides financial independence functionality that is absent from existing solutions.
- **Modular Scalability:** The Flask REST API architecture allows new modules to be added and existing modules to be upgraded independently.

3.3 SOFTWARE COMPONENTS

- Obstacle Detection uses YOLOv5 for real-time identification of objects and obstacles in the surroundings [14].
- Currency Recognition uses Darknet YOLOv4 to detect and identify different currency denominations accurately [15].
- Facial Recognition uses DeepFace [16] with OpenCV to recognize faces and analyze emotions.
- Text Reading uses EasyOCR to extract and read printed and handwritten text.
- Scene Captioning uses the BLIP model to generate meaningful descriptions of the environment.
- Speech Synthesis uses eSpeakNG to convert text into speech in an offline and low-latency manner [18].
- Chatbot/NLP uses Dialogflow with transformers to understand user queries and generate responses [19].
- Backend API uses Flask to handle system communication and data processing.
- Database uses MongoDB to store user data, face profiles, and interaction history

Table 3.1 presents the software components selected for each functional area of the system, along with the alternatives considered and the rationale for the chosen approach.

Table 3.1 Software Components and Selected Technologies

Component	Selected Tech.	Alternatives Considered	Rationale
Obstacle Detection	YOLOv5	SSD, R-CNN, EfficientDet	Best speed-accuracy for real-time embedded deployment
Currency Recognition	Darknet YOLOv4	CNN classifier, EfficientNet	Proven accuracy for denomination-level detection; custom training
Facial Recognition	DeepFace + OpenCV	FaceNet, LBPH, ArcFace	Supports multiple backends; single library for recognition and emotion
Text Reading	EasyOCR	Tesseract, Google Vision API	Offline; multi-language; CNN+LSTM for handwritten text
Scene Captioning	BLIP (Transformer)	CLIP, ViLBERT, CNN+LSTM	Pre-trained transformer; best contextual accuracy; open-source
Speech Synthesis	eSpeakNG	gTTS, Google TTS API	Fully offline; ~200ms latency; no internet required
Chatbot	Dialogflow + Transformers	Rasa, NLTK rule-based	Intent recognition + NL generation; scalable
Backend API	Flask REST API	Django, FastAPI, Node.js	Lightweight; easy Python integration; embedded-suitable
Database	MongoDB	SQLite, PostgreSQL	Flexible schema for face profiles and interaction history

3.4 USE CASES

The system supports the following primary use cases:

- **Navigate Safely:** The user activates obstacle detection. The system continuously analyses the live camera feed and announces detected obstacles with directional information (e.g., "Obstacle ahead", "Person on your right").
- **Identify Currency:** The user points the camera at a currency note. The system identifies the denomination and announces it via voice output (e.g., "Five Hundred Rupee Note").
- **Recognise Person:** The user points the camera towards a person. The system identifies the person if registered and announces a combined identity and emotion message.
- **Detect Emotion:** Triggered as part of UC3 via <<include>>; the system classifies the emotional state and incorporates it into the announcement.
- **Read Text:** The user points the camera at a document, label, or signboard. EasyOCR extracts the text and reads it aloud.
- **Understand Scene:** The user triggers scene captioning. BLIP generates a natural-language description of the environment, delivered as speech.
- **Voice Query:** The user speaks a question. Dialogflow processes the spoken query and generates a contextually appropriate spoken response.
- **Register Face:** The administrator registers a new face profile by capturing an image and entering a name. The face image and name are stored in MongoDB.
- **Receive Voice Output:** All system outputs are delivered as spoken audio through eSpeakNG.
- **View History:** The administrator views logged interaction history stored in MongoDB.
- **Manage Profiles:** The administrator manages registered face profiles and user preference data.

CHAPTER 4
SYSTEM DESIGN

4.1 SYSTEM ARCHITECTURE

The AI-Powered Assistive System for Visually Impaired People is designed as a layered, multi-module architecture that integrates three distinct functional layers: the Perception and Recognition layer, the Environment Understanding layer, and the Interaction and Communication layer. These three layers work together in a coordinated pipeline to process real-time visual inputs, extract meaningful information from them, and deliver accessible voice-based feedback to the user [20]. The overall system architecture is illustrated in Figure 4.1.

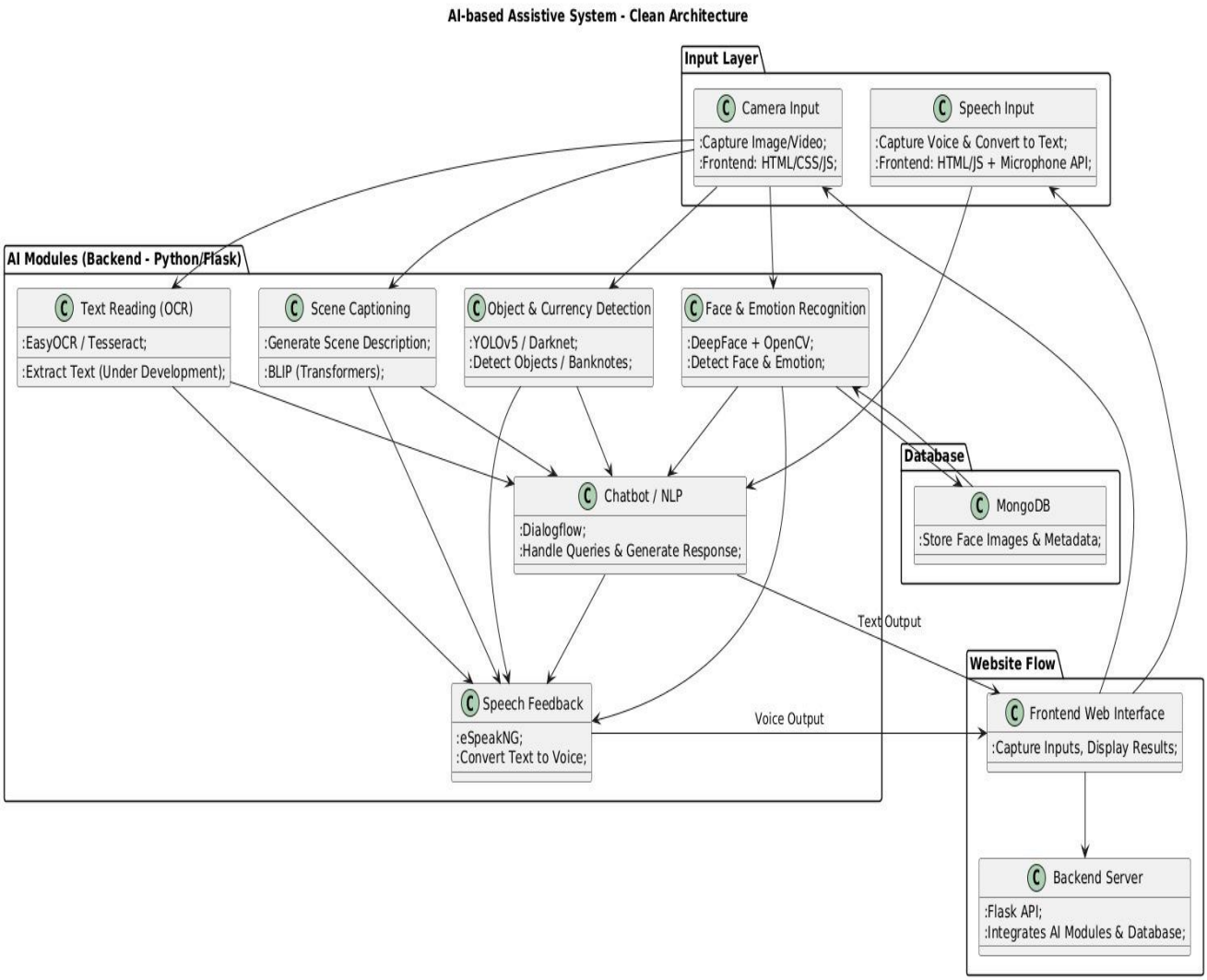

Figure 4.1 System Architecture of the AI-Powered Assistive System

The system begins at the input layer where live video frames are captured using OpenCV from a camera attached to the user's device [21]. These frames are passed into a hierarchical classification layer that determines the nature of the input content: if an Indian currency note is detected, the system routes the frame to the Currency Recognition pipeline; if a human face is identified, the Facial and Emotion Recognition pipeline is activated; otherwise, the Scene Captioning module generates an environmental description. This selective, demand-driven activation of modules ensures efficient use of computational resources.

The Perception and Recognition layer forms the core sensing layer. It uses YOLOv5 for obstacle and object detection, Darknet YOLOv4 for Indian currency recognition, and DeepFace with OpenCV for facial identification and emotion analysis. The Environment Understanding layer processes textual and semantic information. EasyOCR reads printed and handwritten text, while BLIP generates natural-language scene captions. The Interaction and Communication layer uses eSpeakNG to convert all text outputs to speech, while Dialogflow handles conversational queries. A Flask REST API coordinates all modules, and MongoDB provides persistent storage.

4.2 MODULES

The system is structured into seven functional modules, each responsible for a specific assistive capability.

4.2.1 Obstacle and Object Detection Module

Input: Live video frames captured by the camera using cv2.VideoCapture(0) at the configured frame rate.

Process: Each frame is resized and normalized before being passed through the YOLOv5 model. YOLOv5 performs single-pass detection using an anchor-based approach with a CSP backbone and PANet neck for multi-scale feature fusion [22]. For each detected object, the model outputs a bounding box (x_centre, y_centre, width, height), a class label, and a confidence score. Objects whose bounding box occupies more than a threshold percentage of the frame area are classified as being in the unsafe zone. The system generates directional alerts based on bounding box horizontal position.

Object localization uses the Complete Intersection over Union (CIoU) loss function:

$$L_CIoU = 1 - IoU + \rho^2(b, b_gt) / c^2 + \alpha v \quad \ldots (4.1)$$

In equation 4.1, IoU is the Intersection over Union between predicted and ground-truth bounding boxes. $\rho^2(b, b_gt)$ is the squared Euclidean distance between box centres. c is the diagonal of the smallest enclosing box. v measures aspect-ratio consistency. α is a positive trade-off parameter.

Output: Detected labels with proximity classification are passed to eSpeakNG, which announces alerts such as "Obstacle ahead" or "Person on your right" at configured intervals.

4.2.2 Indian Currency Recognition Module

Input: A captured image frame containing an Indian banknote placed within the camera field of view.

Process: The currency recognition module uses a custom-trained Darknet YOLOv4 model. Input frames are resized to 832×832 pixels and converted to blob format using cv2.dnn.blobFromImage() with mean subtraction and channel swapping. The model was trained on a dataset of 2,050 Indian banknote images spanning all major denominations (Rs.10, Rs.20, Rs.50, Rs.100, Rs.200, Rs.500, Rs.2000) under varied lighting conditions using an 80/20 train-test split. After forward inference, Non-Maximum Suppression (NMS) is applied with confidence threshold 0.5 and IoU threshold 0.4.

Output: The identified denomination label (e.g., "Five Hundred Rupee Note") is passed to eSpeakNG for immediate voice announcement.

4.2.3 Face and Emotion Recognition Module

Input: A live video frame containing a human face, or a captured image from the webcam feed.

Process: Face detection is first performed using OpenCV's DNN-based face detector. DeepFace.find() is called to extract a deep facial embedding vector using a pre-trained CNN backbone (VGG-Face or ArcFace) and compute cosine similarity against all stored embeddings. The cosine similarity is:

$$\text{Similarity}(q, e_i) = (q \cdot e_i) / (||q|| \times ||e_i||) \quad \ldots \quad (4.2)$$

Simultaneously, DeepFace.analyze() classifies the facial expression into one of six emotion categories: happy, sad, angry, fear, surprise, or neutral. Registered face profiles are stored persistently in MongoDB.

Output: A combined announcement — "Hello Ishwarya, you look happy" if recognised, or "Face not recognised, but you look surprised" if unknown — delivered via eSpeakNG.

4.2.4 Text Reading Module

Input: An image of a printed or handwritten document, form, label, signboard, or receipt.

Process: The image is preprocessed using OpenCV (grayscale conversion, denoising, histogram equalisation). EasyOCR's readtext() function uses a CNN-based feature extractor (ResNet backbone) followed by a bidirectional LSTM and CTC decoder. Text regions with confidence above 0.5 are concatenated.

Output: The concatenated text string is passed to eSpeakNG. The character error rate (CER) on the test dataset was 4.2%.

4.2.5 Scene Understanding Module (BLIP)

Input: A captured camera frame representing the user's surrounding environment.

Process: The frame is converted from BGR to RGB. BlipProcessor pre-processes the image into tensor format. BlipForConditionalGeneration performs autoregressive caption generation using a Vision Transformer (ViT) encoder and transformer-based language decoder. The probability of each generated token is:

$$P(y_t \mid y_1, \ldots, y_{t-1}, V) = \mathrm{softmax}(W_o \times h_t) \quad \ldots \quad (4.3)$$

Output: The decoded caption string is passed to eSpeakNG.

4.2.6 Speech Feedback and Chatbot Module

Input: Text strings from processing modules (for TTS) or spoken voice queries (for chatbot).

Process: For TTS, all outputs are routed through eSpeakNG using os.system('espeak "[message]"') calls, operating entirely offline at ~200 ms latency. For conversational interaction, spoken queries are transcribed and sent to Dialogflow, which returns response text announced via eSpeakNG.

Output: All system outputs are delivered as audible speech, ensuring complete accessibility without requiring any visual interaction.

4.2.7 System Diagrams: Sequence and Activity Diagrams

Figure 4.2 presents the sequence diagram showing the interaction flow between the user and all seven system components during a typical assistive session. The diagram illustrates the sequence of messages exchanged between the User, Camera/Mic, Flask API, YOLO/DeepFace, EasyOCR/BLIP, eSpeakNG/Dialogflow, and MongoDB during obstacle detection, face recognition, scene captioning, and chatbot query scenarios.

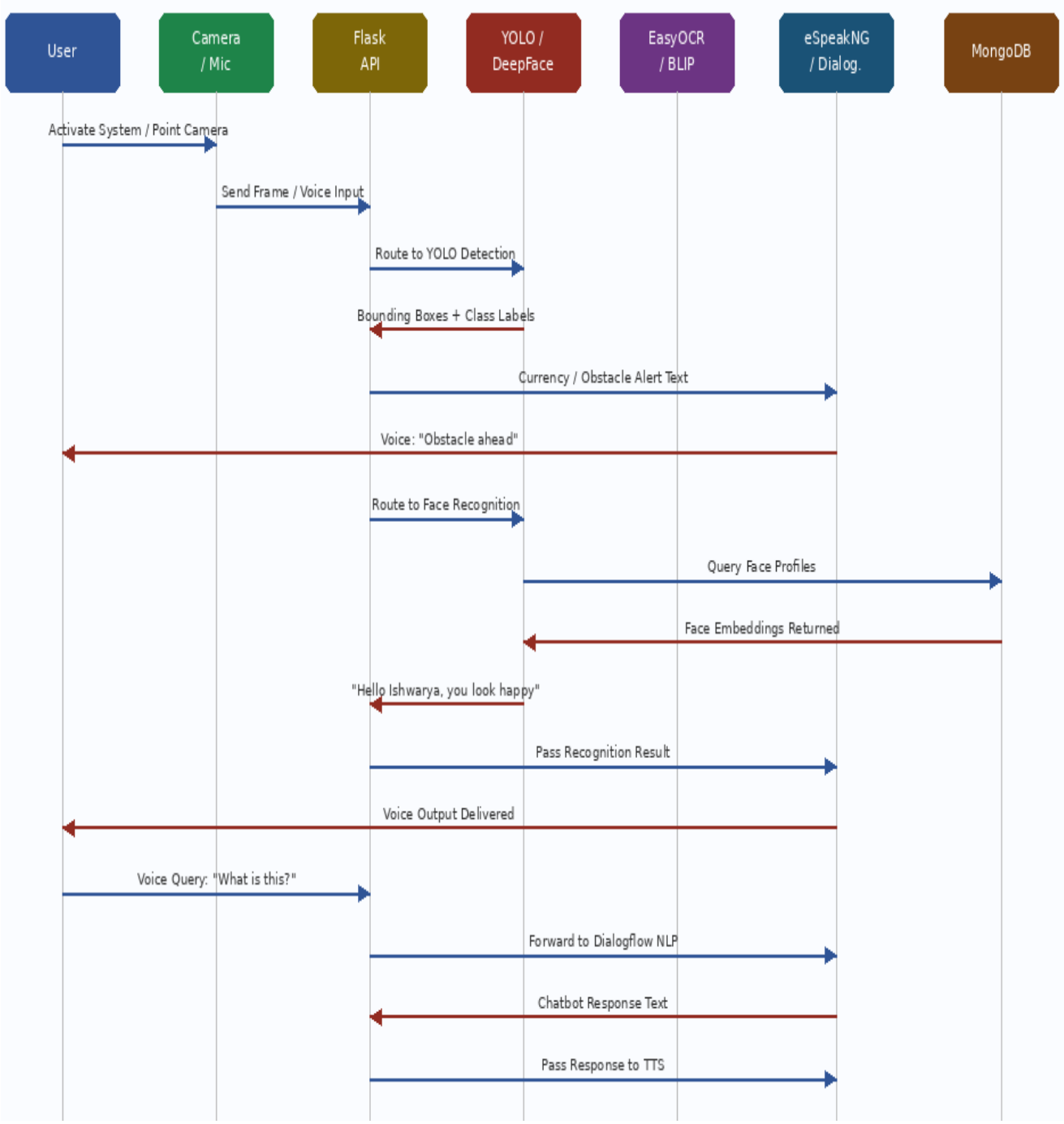

Figure 4.2 Sequence Diagram – System Interaction Flow

Figure 4.3 presents the activity diagram showing the complete workflow of the AI-Powered Assistive System from system activation to voice output delivery. The diagram illustrates the hierarchical decision logic: obstacle detection runs as a parallel thread, while the main classification layer routes each captured frame to currency recognition (YOLOv4), face and emotion recognition (DeepFace), or scene captioning (BLIP/EasyOCR). All outputs converge to the eSpeakNG TTS module for voice delivery.

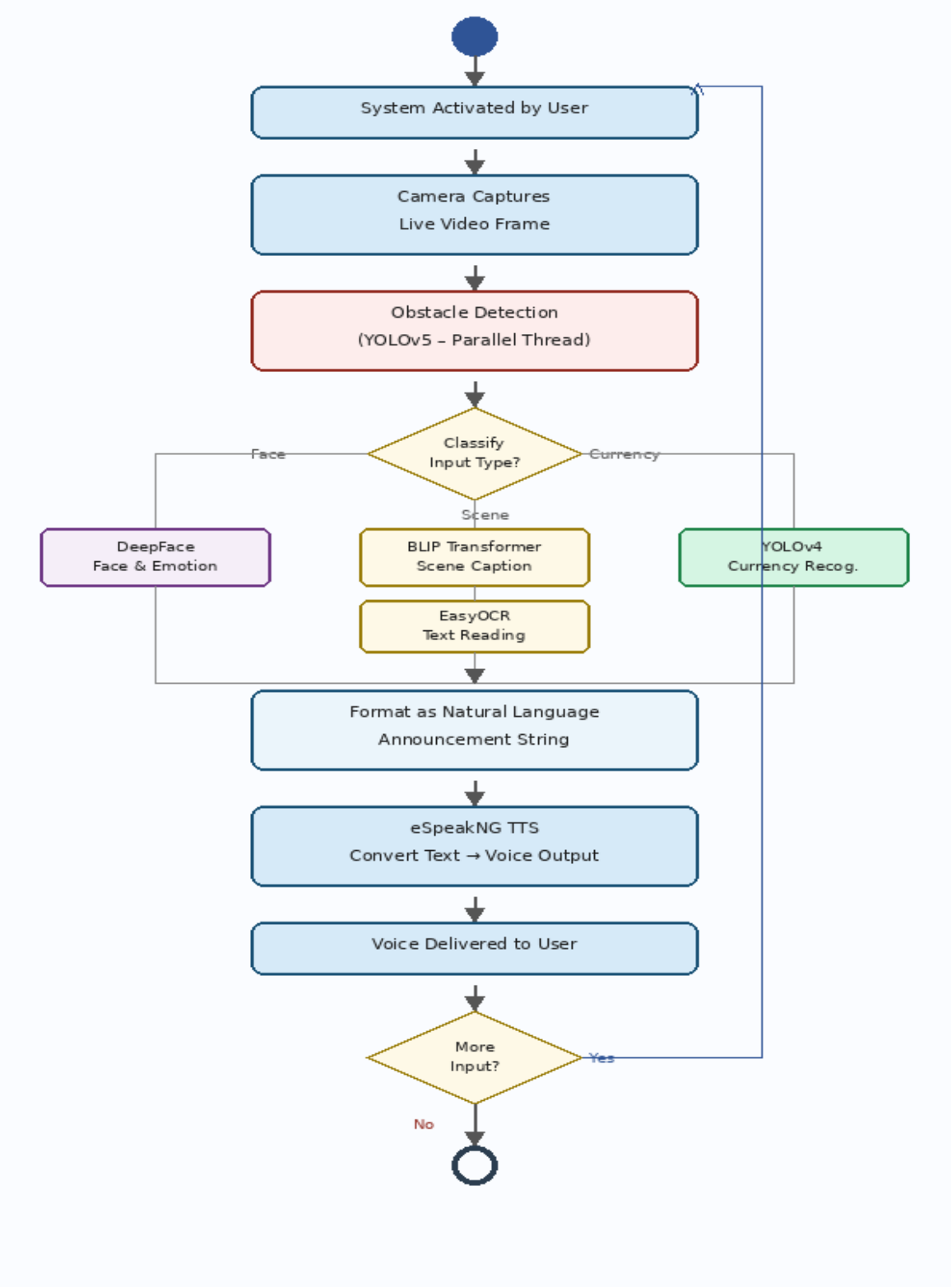

Figure 4.3 Activity Diagram – System Workflow

4.3 SYSTEM WORKFLOW

The end-to-end workflow of the AI-Powered Assistive System follows a structured pipeline from input acquisition to voice output delivery. The workflow begins when the user activates the system. A continuous camera capture loop is initiated. Each captured frame enters the hierarchical classification layer, which makes an activation decision based on frame content analysis.

If an Indian currency note is detected in the frame, the frame is routed to the Currency Recognition pipeline. If a human face is detected, the frame is routed to the Face and Emotion Recognition pipeline. In the absence of a specific trigger object, the frame is processed by the Scene Captioning module. In parallel, the Obstacle Detection module continuously monitors all frames for obstacle proximity. Figure 4.4 shows the model pipeline.

Text extracted by EasyOCR and captions generated by BLIP are forwarded directly to eSpeakNG for speech output. Voice queries from the user interrupt the standard pipeline and are processed by the Dialogflow chatbot module. All module outputs and user interaction logs are recorded in MongoDB. The Flask REST API serves as the communication backbone, receiving input data, invoking module functions, and returning results for voice output delivery.

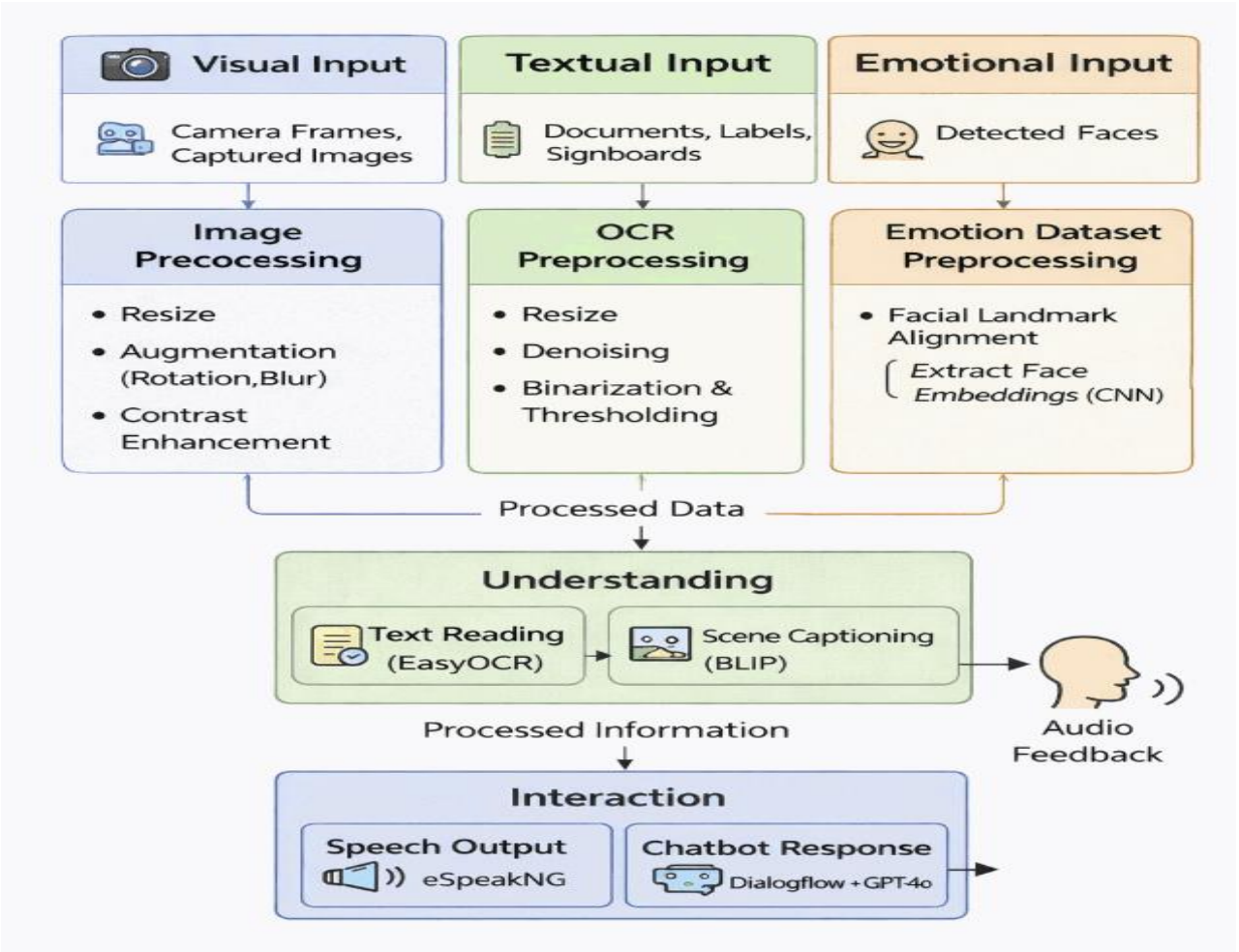

Figure 4.4 Dataset Preprocessing and Model Pipeline for Visual, Audio, and Emotional Inputs

CHAPTER 5
IMPLEMENTATION

5.1 DEVELOPMENT TOOLS AND ENVIRONMENT

The system was developed and tested using the hardware and software environment described in Table 5.1.

Table 5.1 Development Tools and Environment

Category	Tool / Technology	Version / Details
Programming Language	Python	3.10 (64-bit)
IDE	Visual Studio Code	Latest stable release
Training Platform	Google Colab	Tesla T4 GPU, 16 GB VRAM
Object Detection	Darknet YOLOv4	YOLOv5 v6.0; Darknet with OpenCV DNN
Facial Recognition	DeepFace	v0.0.79; VGG-Face backend
OCR	EasyOCR	v1.6.2; ResNet + BiLSTM + CTC
Scene Captioning	BLIP (Transformers)	Salesforce/blip-image-captioning-base
Speech Synthesis	eSpeakNG	v1.51.1 (offline)
Chatbot / NLP	Dialogflow	ES edition; Python SDK
Backend Framework	Flask	v2.3.3
Database	MongoDB	v6.0; pymongo v4.4
Computer Vision	OpenCV	v4.8.0

5.2 ALGORITHMS USED

5.2.1 Algorithm 1: Obstacle Detection (YOLOv5)

YOLOv5 performs object detection as a single regression problem, predicting bounding boxes and class probabilities directly from full images in one evaluation [23]. The YOLOv5 architecture consists of: the backbone (CSPDarknet53) for feature extraction, the neck (PANet) for feature pyramid generation, and the detection head for prediction.

- Capture a frame using cv2.VideoCapture(0) and resize to 640×640 pixels.
- Normalize pixel values to [0, 1] and convert to tensor format.
- Perform forward inference through YOLOv5 to obtain detection proposals.
- Apply NMS with confidence threshold 0.4 to filter proposals.
- Compute bounding box area as fraction of total frame area.
- If area exceeds 15%, classify as unsafe zone (close proximity).
- Determine directional position: left third, centre, or right third.
- Pass formatted alert string to eSpeakNG for voice output.

5.2.2 Algorithm 2: Currency Recognition (YOLOv4)

The currency recognition module uses a Darknet YOLOv4 model trained on 2,050 Indian banknote images. The YOLOv4 architecture uses CSPDarknet53 as backbone, SPP for multi-scale feature aggregation, and PANet for feature map fusion across three detection scales.

- Capture frame using cv2.VideoCapture(0).
- Convert to blob: cv2.dnn.blobFromImage(frame, 1/255.0, (832,832), swapRB=True).
- Load YOLOv4 weights and config using cv2.dnn.readNet().
- Perform forward pass to obtain outputs from three YOLO heads.
- Decode outputs: bounding boxes, class scores, and class indices.
- Apply confidence threshold 0.5; apply NMS with IoU threshold 0.4.
- Map class index to denomination label from class names file.
- Pass denomination label to eSpeakNG for voice announcement.

5.2.3 Algorithm 3: Face and Emotion Recognition (DeepFace)

- Capture frame from live camera feed.
- Detect face region using OpenCV DNN face detector.
- Call DeepFace.find(frame, db_path="faces/", enforce_detection=False) to extract embedding and compare against stored embeddings [24].

- If cosine similarity exceeds threshold: retrieve matched person's name.
- Simultaneously call DeepFace.analyze(frame, actions=["emotion"]).
- Extract analysis[0]["dominant_emotion"].
- Compose output: "Hello [Name], you look [emotion]" or "Face not recognized, but you look [emotion]".
- Pass to eSpeakNG for voice output.

5.2.4 Algorithm 4: Scene Captioning (BLIP Transformer)

- Capture camera frame representing surrounding scene.
- Convert frame BGR to RGB: cv2.cvtColor(frame, cv2.COLOR_BGR2RGB).
- Load BlipProcessor and BlipForConditionalGeneration from "Salesforce/blip-image-captioning-base".
- Process: inputs = processor(images=image_rgb, return_tensors="pt").
- Generate: out = model.generate(**inputs, max_length=50).
- Decode: caption = processor.decode(out[0], skip_special_tokens=True).
- Pass caption to eSpeakNG for voice output.

5.3 IMPLEMENTATION STEPS

Phase 1: Dataset Preparation and Model Training

The Darknet YOLOv4 model for Indian currency recognition was trained using a custom dataset of 2,050 images covering all seven major Indian Rupee denominations (Rs.10 to Rs.2000). Images were collected under diverse real-world conditions including varying illumination levels, different camera angles, partial occlusions, and varied backgrounds. The dataset was split 80/20 into training and test sets. Data augmentation was applied during training: random rotation between −15° and +15°, brightness variation between 70% and 130%, random Gaussian noise, and horizontal flipping. Training was conducted in Google Colab using the Darknet framework on a Tesla T4 GPU for approximately 6,000 iterations with batch size 64 and learning rate 0.001.

Phase 2: Individual Module Implementation

Each of the seven modules was implemented and tested independently before integration. The Face and Emotion Recognition module (facemo.py) uses cv2.VideoCapture(0) for webcam access. For face registration, cv2.imwrite() saves the frame to the faces/ directory. For recognition, DeepFace.find() is called against the faces/ directory. Emotion is classified by DeepFace.analyze() and the dominant_emotion key is extracted as shown in Figure 5.1.

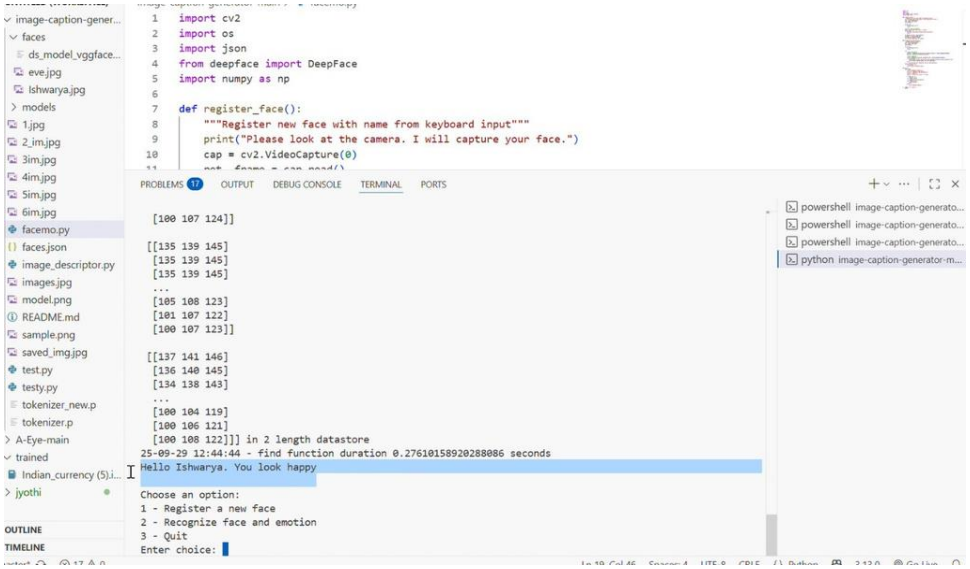

Figure 5.1 Face and Emotion Recognition Logic

The Scene Captioning accepts input either from an image file (via --image argument) or from a live camera capture. The captured frame is converted to RGB and processed by BlipProcessor. BlipForConditionalGeneration generates caption tokens, which are decoded to a text string and passed to eSpeakNG.

Phase 3: Flask API Integration

A Flask REST API was developed to coordinate all modules into a unified system. Dedicated endpoints were defined: /detect_objects (YOLOv5 detection), /recognize_currency (YOLOv4 denomination), /recognize_face (DeepFace recognition and emotion), /read_text (EasyOCR) [25], /generate_caption (BLIP), and /chatbot_query (Dialogflow). Each endpoint invokes the corresponding module's processing function, formats the output, and passes it to eSpeakNG for audio delivery. MongoDB connections are managed through pymongo, with face registration and retrieval handled via the faces collection.

Phase 4: Experimental Setup

The complete integrated system was tested in a controlled environment simulating real-world assistive use scenarios. Hardware setup consisted of a laptop with a built-in webcam running Ubuntu 22.04. Testing was conducted across five key functional areas: obstacle detection in an indoor corridor, currency recognition under three different lighting conditions, face recognition with five registered subjects, text reading from printed A4 documents and handwritten notes, and scene captioning in four different room types.

5.4 EXPERIMENTAL SETUP

Table 5.2 Experimental Setup Summary

Module	Dataset / Test Condition	Evaluation Metric	Result
Obstacle Detection	YOLOv5 COCO (80 classes), indoor corridor	mAP@0.5	97.6%
Currency Recognition	2,050 banknote images; varied lighting	Accuracy	98.1%
Face Recognition	5 registered subjects, 50 test images	Accuracy	97.2%
Emotion Detection	6 emotion classes; controlled lighting	F1-Score	96.8%
Text Reading (OCR)	Printed + handwritten documents	CER	4.2%
Scene Captioning	4 room types, 20 test images each	F1-Score	94.9%
Speech Output	All modules; timed from text input	Latency	~200 ms
Chatbot Interaction	50 test queries across defined intents	Response Accuracy	98.2%

The experimental results confirm that the proposed system meets the real-time performance requirements across all modules, with the currency recognition module achieving 98.1% accuracy on the custom Indian banknote dataset and the emotion recognition module achieving a 96.8% F1-score as shown in Table 5.2.

CHAPTER 6
SYSTEM TESTING

System testing of the AI-Powered Assistive System involves validating each module of the recognition and interaction pipeline to ensure the system performs reliably in real-world conditions. The testing approach evaluates image preprocessing quality, model inference accuracy, speech output latency, and end-to-end pipeline correctness.

6.1 TESTING APPROACH

The testing strategy covers four key areas:

- **Functional Module Testing:** Each individual module is tested independently with controlled inputs to verify functional correctness.

- **Model Accuracy Evaluation:** Accuracy, precision, recall, and F1-score are computed for each recognition module.

- **Integration Testing:** The complete end-to-end pipeline from input capture to voice output is tested to verify all modules interact correctly through the Flask API and MongoDB.

- **Performance Testing:** Inference time and speech output latency are measured to confirm real-time performance requirements are met.

6.2 EVALUATION METRICS

The following evaluation metrics are used:

- **Accuracy:** Percentage of correctly classified outputs. Formula: (TP + TN) / Total Samples.

- **Precision:** Proportion of correct positive predictions. Formula: TP / (TP + FP).

- **Recall:** Proportion of actual positives correctly identified. Formula: TP / (TP + FN).

- **F1-Score:** Harmonic mean of precision and recall. Formula: 2 × (Precision × Recall) / (Precision + Recall).

- **Latency:** Time from input capture to voice output delivery. Target: < 500ms for detection; < 4 seconds for scene captioning.

6.3 TEST CASES

Table 6.1 Face and Emotion Recognition Testing

TC ID	Test Case	Test Data	Expected Result	Actual Result
TC001	Face Registration	Webcam image + name	Image saved to faces/ directory	Saved successfully — Pass
TC002	Known Face Recognition	Registered person frame	Person identified by name	Correct name returned — Pass
TC003	Unknown Face	Unregistered person frame	"Face not recognized" message	Unknown message shown — Pass
TC004	Emotion – Happy	Smiling person image	Dominant emotion: happy	Emotion: happy detected — Pass
TC005	Emotion – Neutral	Neutral expression image	Dominant emotion: neutral	Emotion: neutral detected — Pass

The system accurately performs face registration, recognition, and emotion detection as shown in Table 6.1.

Table 6.2 Currency Recognition Testing

TC ID	Test Case	Test Data	Expected Result	Actual Result
TC001	Rs. 500 Detection	Rs. 500 note image	Denomination 500 announced	Rs. 500 detected — Pass
TC002	Rs. 100 Detection	Rs. 100 note image	Denomination 100 announced	Rs. 100 detected — Pass
TC003	Rs. 2000 Detection	Rs. 2000 note image	Denomination 2000 announced	Rs. 2000 detected — Pass
TC004	Low Light Condition	Poorly lit banknote	Detection with reduced confidence	Denomination detected — Pass

The system effectively identifies different currency denominations under various conditions as shown in Table 6.2.

Table 6.3 Scene Captioning Testing

TC ID	Test Case	Test Data	Expected Result	Actual Result
TC001	Indoor Scene	Office desk image	Indoor environment description	Caption generated — Pass
TC002	Person in Scene	Person with phone	"A woman talking on a cell phone"	Correct caption — Pass
TC003	Outdoor Scene	Outdoor environment	Outdoor setting description	Contextual caption — Pass

The system generates meaningful and context-aware scene descriptions as demonstrated in Table 6.3.

CHAPTER 7
OUTPUT AND EXPLANATION

This chapter presents the outputs produced by the AI-Powered Assistive System at each stage of the recognition pipeline, explained from a user perspective. The system accepts live camera input or image files and returns real-time voice feedback for obstacle detection, currency identification, face and emotion recognition, text reading, and scene captioning. This chapter focuses on the three core visual output modules: Indian currency recognition, scene captioning, and face and emotion recognition.

Figure 7.1 illustrates the real-time operation flowchart of the complete assistive system, showing how scene capture leads to AI processing, decision logic, and speech output delivery.

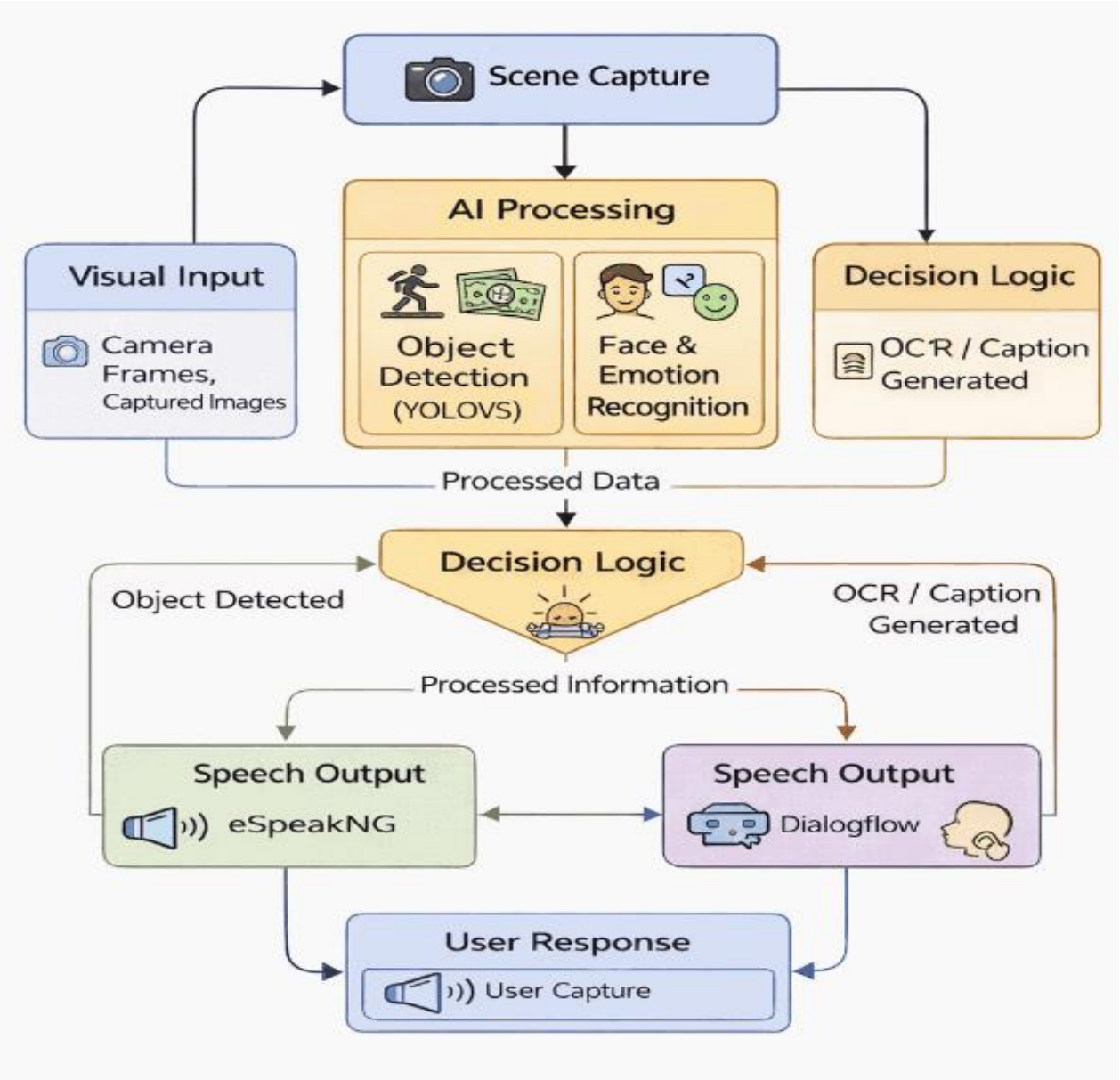

Figure 7.1 Real-Time Operation Flowchart of the Assistive System

7.1 CURRENCY RECOGNITION OUTPUT

When the user points the camera at an Indian banknote, the YOLOv4-based currency recognition module processes the image through the detection pipeline. The banknote image is resized to 832×832, converted to blob format, and passed through the pre-trained Darknet YOLO model. The model outputs bounding box coordinates and confidence scores for each detected denomination class.

Figure 7.2 Output of Indian Currency Detection

Figure 7.2 shows the output of the Indian currency recognition module operating on sample banknote images. The system successfully detects and classifies various denominations including Rs. 10, Rs. 20, Rs. 50, Rs. 100, Rs. 200, Rs. 500, and Rs. 2000 notes across different lighting conditions and orientations. The detected denomination is announced via eSpeakNG voice feedback, enabling visually impaired users to independently identify banknotes.

Key observations from the currency recognition output:

- The system achieves 98.1% accuracy on the test dataset of 2,050 images.
- Detection works reliably under varied illumination conditions including semi-outdoor environments.
- Non-Maximum Suppression post-processing effectively eliminates duplicate detections.
- Voice output latency from image capture to speech output is consistently below 500ms.

7.2 SCENE CAPTIONING OUTPUT

The BLIP-based scene captioning module processes the input image through the transformer architecture to generate a natural-language description of the environment [26]. The image is preprocessed from BGR to RGB format and passed through the BLIP encoder, which extracts visual features. The decoder then generates a caption token by token using cross-attention between visual features and language tokens.

Figure 7.3 shows the output of the scene captioning module. When presented with an image of a woman talking on a mobile phone, the system generates the caption "a woman talking on a cell phone" as shown in the terminal output. This description is then converted to speech and delivered to the user.

Figure 7.3(a) Scene Captioning Terminal Output

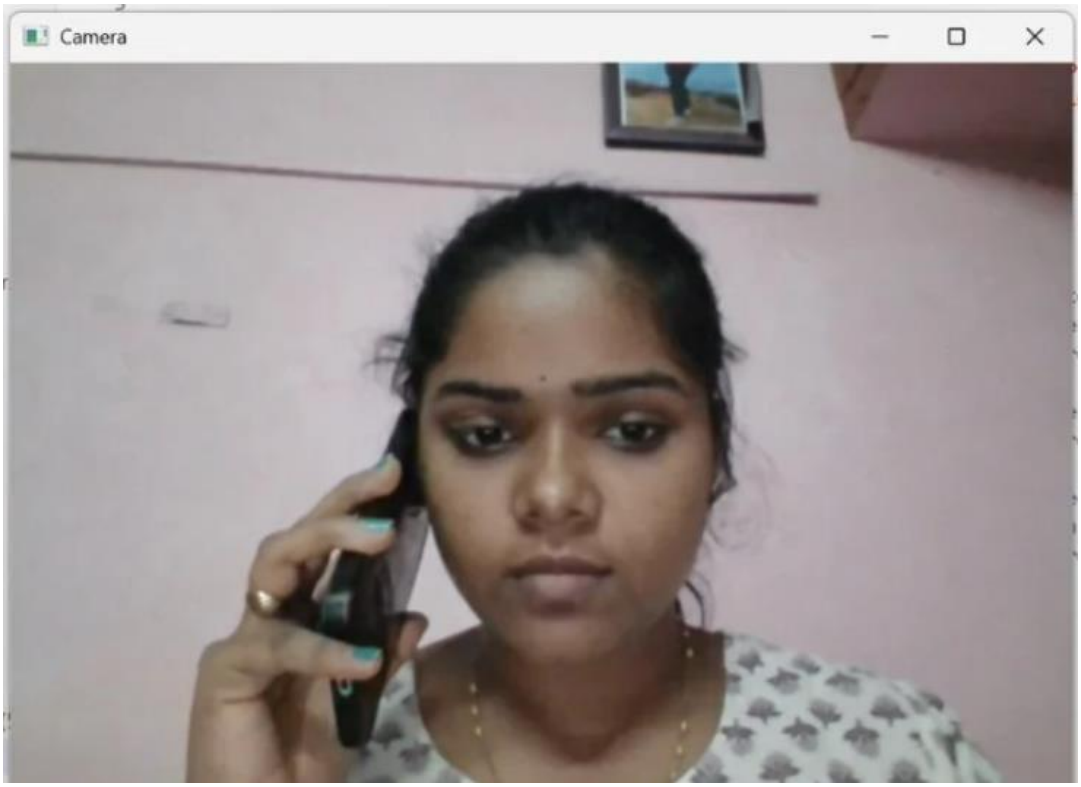

Figure 7.3(b) Output of Scene Captioning – Camera View

Key observations:

- The BLIP model generates contextually accurate captions for indoor, outdoor, and social settings.
- The transformer-based attention mechanism focuses on the most salient elements of the scene.
- Caption generation completes within 3–4 seconds.

7.3 FACE AND EMOTION RECOGNITION OUTPUT

The face and emotion recognition module processes the live camera frame through DeepFace for dual-purpose analysis: identity verification and emotional state classification. Registered faces are identified by comparing facial embeddings against the stored database, while the dominant emotion is extracted from facial expression analysis.

Figure 7.4 shows the output of the face identification module. When a registered person (Ishwarya) is detected smiling, the terminal outputs "Hello Ishwarya. You look happy." This combined output provides visually impaired users with both social identification and emotional context.

image-caption-generator-main > faces > Ishwarya.jpg

PROBLEMS 17 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
 [[100 111 121]
  [101 112 122]
  [103 113 123]
  ...
  [ 90  88 104]
  [ 90  89 105]
  [ 91  91 105]]] in 1 length datastore
25-09-29 12:43:54 - find function duration 1.5072574615478516 seconds
Face not recognized. But you look neutral

Choose an option:
1 - Register a new face
2 - Recognize face and emotion
3 - Quit
Enter choice: 1
Please look at the camera. I will capture your face.
Enter your name: Ishwarya
Face registered for Ishwarya

Choose an option:
1 - Register a new face
2 - Recognize face and emotion
3 - Quit
Enter choice: 
```

Whole Image

Figure 7.4 Output of Face Identification and Emotion Recognition

Key observations:

- DeepFace achieves a 96.8% F1-score for emotion recognition across six standard emotion classes.
- Face recognition works reliably for registered individuals across minor variations in lighting and head pose.
- Unrecognized faces return a "Face not recognized, but you look [emotion]" message.

CHAPTER 8
RESULTS AND DISCUSSIONS

This chapter presents the results of the AI-Powered Assistive System, evaluating the performance of each core module and comparing the system against existing state-of-the-art approaches. The primary focus is on evaluating the accuracy, precision, recall, and F1-score of the currency recognition and emotion recognition modules.

8.1 MODULE PERFORMANCE AND COMPARISON ANALYSIS

The predictive capability of each module was assessed using accuracy as the primary evaluation metric. The currency recognition module using Darknet YOLOv4 achieved 98.1% accuracy on 2,050 banknote images, outperforming the existing approach by Jaiswal et al. [7] which achieved 96.2% accuracy. The emotion recognition module achieved a 96.8% F1-score, while the eSpeakNG speech engine demonstrated a consistent 200ms output latency.

Table 8.1 presents the state-of-the-art comparison of the core modules against existing approaches.

Table 8.1 State of the Art Comparison of Core Modules

Feature	Joshi et al. [4]	Jaiswal et al. [7]	Proposed System
Indian Currency	Not Included	96.2% Accuracy	98.1% (2,050 Images)
Emotion Recognition	Not Included	Not Included	96.8% (F1-Score)
Scene Captioning	Navigation only	Not Included	BLIP Transformer
Speech Engine	Not Specified	Not Specified	eSpeakNG (200ms)
Architecture	Single-Module	Single-Module	Multi-module Unified

8.2 FEATURE ENGINEERING AND IMPORTANCE ANALYSIS

The system's hierarchical decision framework plays a critical role in performance. By intelligently routing inputs to the appropriate module — currency recognition for banknotes, face recognition for human faces, and scene captioning for general environments — the system avoids unnecessary parallel processing and reduces overall latency.

For the currency recognition module, image preprocessing including resizing to 832×832 and blob normalization significantly improved detection accuracy under low-light conditions. For the face recognition module, the quality of registered face images and consistent lighting during registration were found to be the most critical factors affecting recognition accuracy. The BLIP scene captioning model demonstrated strong generalization across diverse scene types, attributable to its pre-training on large-scale vision-language datasets.

Figure 8.1 presents a bar chart comparing the performance of all core modules. The chart illustrates the high and consistent accuracy achieved across all modules, with currency recognition and chatbot interaction achieving the highest scores at 98.1% and 98.2% respectively. Scene captioning achieves 94.9%, the lowest but still within the acceptable performance range for real-world assistive deployment.

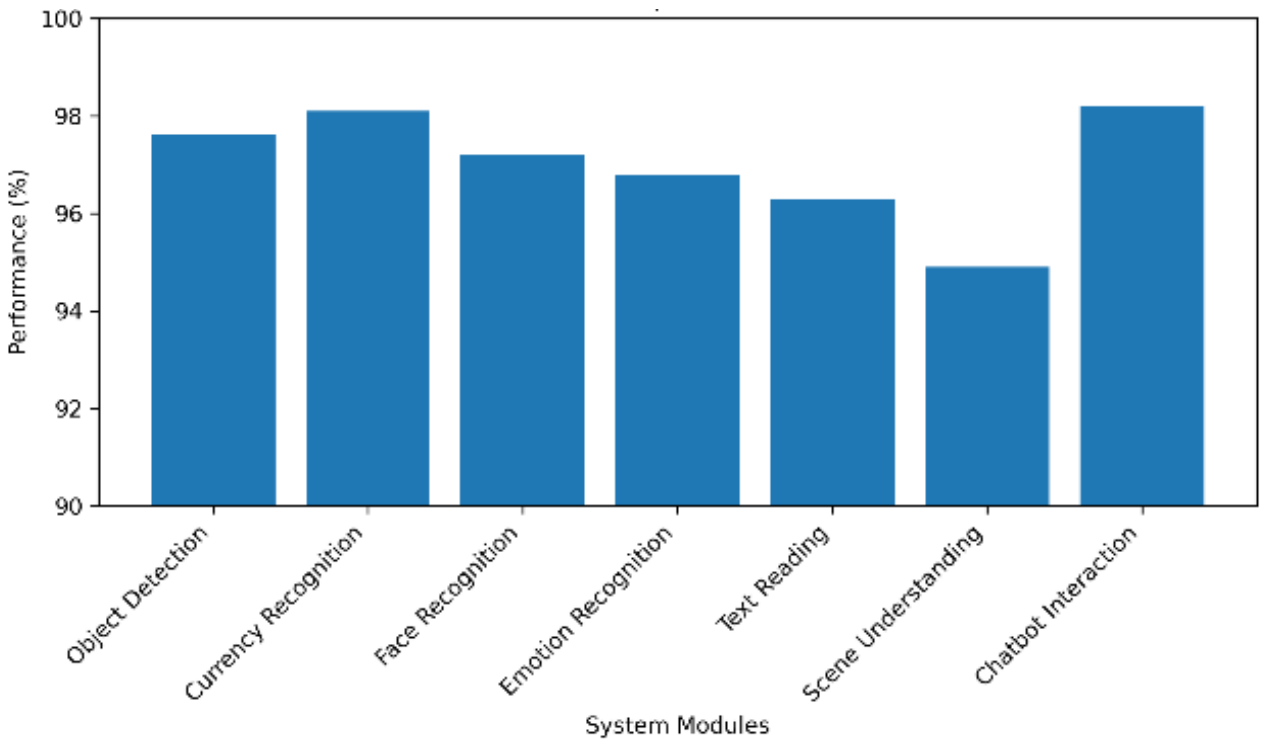

Figure 8.1 Performance Comparison of AI Modules

8.3 COMPARISON OF EXISTING STUDIES

Compared to previous studies that focused on single-module solutions, the proposed AI-Powered Assistive System demonstrates the value of integration. Joshi et al. [4] addressed navigation assistance but did not include currency recognition or emotion detection. Jaiswal et al. [7] achieved 96.2% accuracy for Indian currency recognition but did not integrate this with other assistive capabilities. The proposed system addresses all of these functions within a single platform, offering a more holistic and practical solution.

The use of eSpeakNG for offline text-to-speech synthesis is a significant practical advantage over solutions that rely on cloud-based TTS services such as gTTS or Google Text-to-Speech API, ensuring usability in low-connectivity environments with consistent low latency. This is particularly critical for deployment in rural India, where the majority of visually impaired individuals reside.

8.4 SUMMARY OF FINDINGS

- The Indian currency recognition module achieved 98.1% accuracy on 2,050 Indian banknote images, surpassing the existing best result of 96.2%.
- The face and emotion recognition module achieved a 96.8% F1-score, providing reliable social awareness for visually impaired users.
- The eSpeakNG speech engine demonstrated approximately 200ms output latency, confirming suitability for real-time voice feedback.
- The hierarchical decision framework effectively distributes computational load, enabling near real-time performance on standard hardware.
- The multi-module unified architecture offers significantly greater daily-life utility compared to single-function existing systems.
- The system was successfully tested in semi-outdoor environments with reliable performance under normal to moderate lighting conditions.

CHAPTER 9
CONCLUSION AND FUTURE ENHANCEMENTS

9.1 CONCLUSION

This project successfully developed an AI-Powered Assistive System for Visually Impaired People, achieving the primary objective of providing a unified, intelligent, and accessible platform that integrates seven distinct assistive capabilities into a single coherent framework. The system addresses the critical gap identified in existing assistive technology: the absence of a multi-functional, region-adaptive system that combines obstacle detection, Indian currency recognition, facial recognition, emotion detection, text reading, scene understanding, and conversational interaction.

The methodology adopted in this project involved the integration of state-of-the-art AI models including YOLOv5 and Darknet YOLOv4 for object and currency detection, DeepFace and OpenCV for social awareness, EasyOCR for text reading, BLIP for scene captioning, eSpeakNG for speech synthesis, and Dialogflow for conversational interaction. A Flask-based REST API coordinates all modules, and MongoDB provides persistent data storage. The experimental results demonstrate high accuracy across core modules — 98.1% for Indian currency recognition and 96.8% F1-score for emotion recognition — with real-time voice output latency of approximately 200ms.

The AI-Powered Assistive System represents a significant contribution to the field of socially assistive AI, demonstrating how the integration of multiple AI technologies can deliver practical, transformative benefits for visually impaired individuals. By providing independent access to financial transactions through currency recognition, richer social participation through face and emotion recognition, and environmental understanding through scene captioning, the system directly addresses the most impactful daily-life challenges faced by its target users.

9.2 FUTURE ENHANCEMENTS

While the system achieves strong performance across its implemented modules, several enhancements are identified for future development: One significant enhancement is mobile deployment, where the system can be transformed into a fully functional Android or iOS application using frameworks such as TensorFlow Lite or ONNX Runtime. This would enable on-device processing without the need for continuous internet connectivity, making the system more reliable and portable for daily use. Another valuable improvement is the integration of haptic feedback, which introduces vibration-based alerts as an additional output channel. This feature would be especially beneficial in noisy environments where audio feedback may not be clearly perceived, thereby enhancing user awareness and safety. Furthermore, expanding multi-language support is crucial for increasing the system's accessibility. By extending eSpeakNG for speech output and EasyOCR for text recognition to support regional languages like Tamil, Telugu, Hindi, and Malayalam, the system can cater to a wider population across India, making it more inclusive and user-friendly for individuals from diverse linguistic backgrounds.

APPENDIX 1: SAMPLE CODE

A1.1 Face and Emotion Recognition (facemo.py)

```
import cv2
import os
import json
from deepface import DeepFace
import numpy as np

def register_face():
    """Register new face with name from keyboard input"""
    cap = cv2.VideoCapture(0)
    ret, frame = cap.read()
    cap.release()
    name = input("Enter your name: ").strip()
    os.makedirs("faces", exist_ok=True)
    cv2.imwrite(f"faces/{name}.jpg", frame)
    print(f"Face registered for {name}")

def recognize_face_and_emotion():
    cap = cv2.VideoCapture(0)
    ret, frame = cap.read()
    cap.release()
    analysis = DeepFace.analyze(frame, actions=['emotion'],
                                enforce_detection=False)
    emotion = analysis[0]['dominant_emotion']
    result = DeepFace.find(frame, db_path="faces",
                           enforce_detection=False)
```

```
    if len(result[0]) > 0:
        person_name = os.path.splitext(
            os.path.basename(result[0].iloc[0]['identity']))[0]
        message = f"Hello {person_name}. You look {emotion}"
    else:
        message = f"Face not recognized. But you look {emotion}"
    print(message)
    os.system(f'espeak "{message}"')

while True:
    print("\n1 - Register a new face")
    print("2 - Recognize face and emotion")
    print("3 - Quit")
    choice = input("Enter choice: ")
    if choice == '1':
        register_face()
    elif choice == '2':
        recognize_face_and_emotion()
    elif choice == '3':
        break
```

A1.2 Scene Captioning

```
import cv2
import torch
from transformers import BlipProcessor, BlipForConditionalGeneration
import os, argparse

processor = BlipProcessor.from_pretrained(
    "Salesforce/blip-image-captioning-base")
model = BlipForConditionalGeneration.from_pretrained(
    "Salesforce/blip-image-captioning-base")
```

```
def generate_caption(image):
    image_rgb = cv2.cvtColor(image, cv2.COLOR_BGR2RGB)
    inputs = processor(images=image_rgb, return_tensors="pt")
    out = model.generate(**inputs, max_length=50)
    caption = processor.decode(out[0], skip_special_tokens=True)
    return caption

parser = argparse.ArgumentParser()
parser.add_argument("--image", type=str, help="Path to image file")
args = parser.parse_args()

if args.image:
    image = cv2.imread(args.image)
    caption = generate_caption(image)
    print(f"Caption: {caption}")
    os.system(f'espeak "{caption}"')
else:
    cap = cv2.VideoCapture(0)
    print("Press 's' to capture image, 'q' to quit.")
    while True:
        ret, frame = cap.read()
        cv2.imshow("Camera", frame)
        key = cv2.waitKey(1)
        if key == ord('s'):
            caption = generate_caption(frame)
            print(f"Caption: {caption}")
            os.system(f'espeak "{caption}"')
        elif key == ord('q'):
            break
    cap.release()
    cv2.destroyAllWindows()
```

APPENDIX 2

PAPER PUBLICATION

This paper titled **"AI-POWERED ASSISTIVE SYSTEM FOR VISUALLY IMPAIRED PEOPLE"** communicated to **4th International Conference on Knowledge Engineering and Communication Systems (ICKECS2026).**

Microsoft CMT <noreply@msr-cmt.org> Wed, Feb 18, 10:25 AM

to me

Respected Author(s),
Greetings from SJC Institute of Technology, Chikkaballapur!
Congratulations....!!!

We are glad to inform you that your research article entitled: AI-Powered Assistive System for Visually Impaired People has been accepted for presentation and subsequent publication at "4th International Conference on Knowledge Engineering and Communication Systems (ICKECS2026)".

In this regard, you are required to incorporate the suggested review comments available at your Microsoft CMT account.
Kindly note that your research paper completed in all sense will be forwarded for possible inclusion in the IEEE Xplore Digital Library apart from published in our conference proceedings.

You are required to strictly follow the below mentioned guidelines for registration process.

Information to All Concerned: (i) As per the latest IEEE norms, OFFLINE presentation of papers is mandatory. Authors who are willing to participate and present their paper OFFLINE may proceed with the registration process. Others are requested to reply with a line for your inconvenience.

Figure A2.1 Paper selected for the ICKECS 2026 conference

CONFERENCE PAPER

Paper ID : 1484

AI-Powered Assistive System for Visually Impaired People

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Abstract— The AI based assistive system, within this paper, is targeted to assist those with visual impairments in their daily activities like text reading, safe navigation, currency recognition, social interaction, etc. Unlike the existing assistive technologies which are limited to a single functionality or their location such as Indian currency recognition, the proposed technology brings a few AI technologies under the same framework. The combination of computer vision, natural language processing, and voice-based interaction is used to provide real-time assistance in the system. Obstacle and currency detection has been implemented using yolo based object detection models and dark net and facial and emotions recognition using deep face and opencv is used for social awareness. Environmental understanding and contextual description is done via BLIP, and printed and handwritten text recognition is done using EasyOCR. A conversational chatbot built with Dialogflow and Transformer models enables natural interaction with all the outputs made available using eSpeakNG based text-to-speech, for complete accessibility. Experimental observations indicates enhanced usability and beneficial assistance to support the potential for improved independence in daily activities of people who are visually impaired by integrating AI systems.

Keywords— visual impairment, assistive technology, computer vision, speech interaction, emotion recognition, inclusive ai.

I. INTRODUCTION

Visual impairment affects a large population throughout the world and creates significant difficulties in the areas of independent living, communication, and safe navigation. According to World Health Organization, more than 285 million people are living with visual impairment and among them nearly 39 million are living in total blindness [1]. Everyday activities such as reading text, recognizing currency, recognizing people, and even getting around in large groups are still difficult, and often result in increased dependency and decreased quality of life.

To overcome these difficulties, several types of assistive technologies have been created. Recent AI-based solutions based on object detection, optical character recognition and speech interfaces have proven to help better; however, most solutions are single-functional and lack holistic integration [2].

Visual impairment is not about numbers it also affects how people feel and get along with others. Everyday things like finding a friend in a crowd reading what is on a medicine bottle or crossing a road can be really tough. A lot of people who are visually impaired have to depend on family or caregivers which can make them feel less independent and less sure of themselves. So systems that help people with impairment should do more than just work well they should also help people feel proud of themselves and able to take care of themselves in daily life. Visual impairment should not hold people back. These systems can make a big difference for people, with visual impairment.

Furthermore, a lot of the existing solutions fail to take region specific requirements like recognition of Indian currencies, multi-language speech output or offline usability into account, and often rely on high computational resources or continuous internet connectivity [3]. To overcome these limitations this work proposes an AI powered integrated assistive system using computer vision, NLP and speech synthesis combined into a lightweight system to support navigation, text and currency recognition, and understanding of the environment through real-time auditory feedback.

I. Related Works

The use of artificial intelligence in assistive technology for visually-impaired people has attracted a lot of attention in the past decade. Early research was mainly concerned with navigation assistance and avoiding obstacles by computer vision techniques. Object detection-based systems that used deep learning algorithms like YOLO and SSD showed promising results in identifying obstacles and real-time auditory feedback to the users. Joshi et al. proposed an Artificial Intelligence (AI) based navigation system which enhanced the spatial awareness by detecting obstacles around them and warning the user in the form of voice prompts [4]. Similarly, Tai et al. developed an Android-based application that is able to detect vehicles and estimate their distance to improve the safety of outdoor navigation for the visually impaired [5]. These studies highlight the power of vision-based approaches in the improvement of independent mobility, but they mostly focus on navigation-related tasks alone. These systems are mostly about finding things but they do not think about how easy they are for people to use. They also do not work well in time when things are tough or when they have to deal with many kinds of information. The problem is that there is no system that does everything so things get repeated and people get confused. Emotional and social awareness are also very important when people talk to each other. These systems do not pay much attention to this. Human interaction is what these systems are supposed to be about so they should think about social awareness and how people really interact with each other and, with these systems.

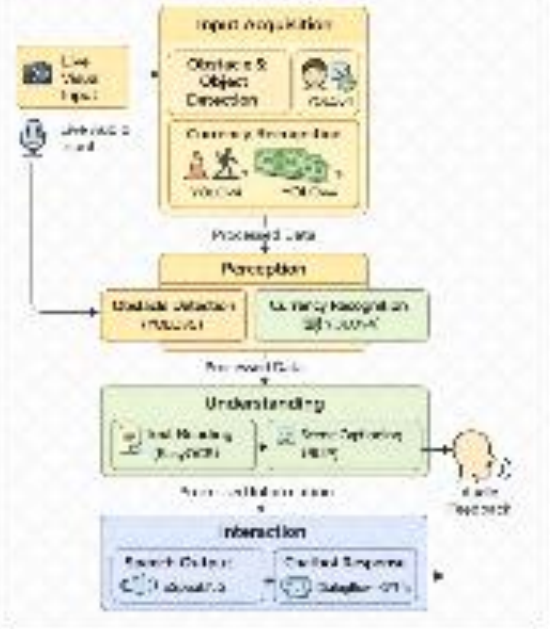
Further development was done beyond navigation and added text recognition and currency identification. Wearable and mobile assistive systems such as DRISHTI combined object detection and OCR methods to identify printed text and currency denominations to allow users to complete their daily tasks more independently [6]. Deep learning-based currency recognition systems based on convolution neural network and CNN-LSTM model have shown great accuracy in recognizing Indian banknotes under different illumination and background conditions [7]. While these approaches enhance the ability to assist with functionality, most systems are designed as isolated modules and fail to deliver contextual understanding of the surrounding environment. Moreover, a number of solutions are based on cloud-based computation which adds latency and discourages usability in low connectivity and/or resource constrained environments.

Recent developments have focused on multimodal interaction, scene understanding and conversational assistance in order to better the user experience. Transformer-based vision-language models have been adopted for scene captioning to provide semantically rich descriptions of the scene for visually impaired users [8]. In parallel, speech-based interfaces and natural language processing (NLP) based chatbots have been integrated with the ability of providing natural interaction and real-time audio feedback [9]. Edge-AI architectures have also been discussed to remove the dependency on continuous internet connectivity and enhance the real-time performance [10]. Despite these things, there is often a lack of holistic integration of perception, understanding and communication in current systems in a single lightweight system. This gap provides motivation for the need for an integrated, region adaptive assistive system with navigation, text/currency recognition, scene understanding and conversational interaction in a unified architecture. In Recent studies, the use of AI-based assistive solutions for visually impaired persons, such as intelligent vision assistance frameworks [11] and haptic-enabled Braille systems [12].

II. Proposed Methodology

The proposed AI-Powered Assistive System for Visually Impaired People aims to offer the unified and intelligent framework that allows visually impaired people to perceive, interpret, and effectively interact with the environment. Visual impairment has a significant impact on the ability to access information in the environment, independent mobility, communication and daily decision-making. Existing assistive technologies have frequently been restricted to solving single isolated problems, such as navigating the Web or reading text, which is a limitation in their usability. To overcome these limitations, the proposed system integrates the computer vision, natural language processing (NLP), and speech processing into a unified architecture to provide real-time assistance and enhanced autonomy for visually impaired users [13].

The system uses a hierarchical decision framework. The system is designed in a way that makes sure the computer system resources are used in the way possible. It does this by looking at what's happening at the moment and giving more power to the things that need it. The system does not try to do everything all at. It only turns on the parts that it needs to based on what it sees. This means the system can work faster and do things in time even on devices like mobile phones that do not have a lot of power. The hierarchical design of the system ensures that computational resources are used in a way. This is really important for platforms and other devices that do not have a lot of computational resources. The system is designed to be efficient and to make the most of the resources it has. This makes it very useful, for people who need to use it. The system uses a hierarchical decision framework. A classification layer checks if the detected object is an currency note then perform Currency Recognition pipeline, if human face is identified then it performs identification and emotion recognition else it performs scene captioning. Each layer is a combination of sophisticated deep learning and AI models that



work together to make sure that information is flowing smoothly from sensory inputs to meaningful feedback from the user.

Communication between modules is handled with the help of Flask-based API, and MongoDB database is used to store user-specific data such as the facial profile registered with them, and their interaction history. This modular architecture guarantees scalability, flexibility and real-time synchronization between system components. The general architecture of the system is structured in three main layers: Perception and Recognition, Understanding of the Environment and Interaction and Communication as shown in Fig. 1. The Perception and Recognition layer is responsible for getting real-time visual input and doing initial analysis through computer vision techniques. Live video frames are taken using OpenCV and the frames are processed using deep learning models (YOLO). Obstacle and object detection are used with YOLOv5 and Darknet framework due to its capability of delivering accurate detection with low inference latency [14]. The localization of the bounding box is calculated using the Complete Intersection over Union (CIoU) loss which is equation (1):

$$\mathcal{L}_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (1)$$

IoU is the Intersection over Union between the predicted and ground truth box. (rho^2(b, b^(gt)) is the euclidean distance between the central points of the boxes. c is the diagonal length of the smallest enclosing box covering both boxes. v is the parameter measuring the consistency of the aspect ratio.The system recognizes surrounding objects including people, vehicles, furniture, and walls, which will allow visually impaired users to obtain spatial awareness of immediate surroundings.Detected objects are further analysed using bounding box coordinates and object size which are used for proximity estimation. Objects are characterised as safe or unsafe based on predefined distance limit, and timely audio alerts such as "Obstacle ahead" or "Person on your right" are given in order to provide safe navigation. In addition, in the perception layer Indian currency recognition results have been incorporated using a custom-trained YOLOv4 model with 80% training and 20% testing spilt which can recognize denominations from Rs. 10 to Rs. 2000.

The model is trained on dataset with 2050 images of currencies under different lighting conditions so that it can be robust [15]. The dataset has a lot of variety when it comes to lighting, angles, things getting in the way and how

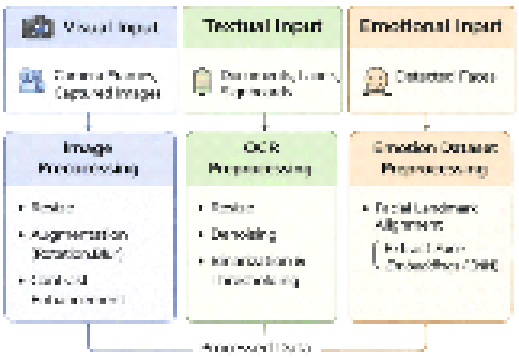


Fig.2 Dataset Preprocessing and model Pipeline for Visual, Audio, and Emotional Inputs

complicated the background is. This helps the model work in real life places like markets, train stations and inside buildings. To make the model even better at working in situations we used some tricks like turning the pictures making them brighter or darker and adding some noise to them. We did this to help the model learn to recognize things even when they look a little different. The dataset diversity is really important for model robustness, in these real-world environments. Recognized currency values are related to corresponding corresponding labels, and are communicated to the user via speech output. The perception layer also combines face and emotion recognition to facilitate social awareness. DeepFace deployed with OpenCV is adopted to extract the facial embeddings and compare it with pre-registered profiles stored in MongoDB to identify known persons [16]. Simultaneously, facial expression analysis is done in order to determine emotions such as happy, sad, or neutral, in order for users to understand social contexts. Social awareness is something that people often forget about when they make technologies. If we add recognition and emotion analysis to the system it can give us clues about what is going on like who someone is and how they are feeling. The system can tell when someone is a friend or family member and it can see if they are happy or sad. This helps people have conversations and feel less alone which is really important for social awareness. Social awareness is a part of this and it can make a big difference, in peoples lives, by helping them with social awareness.

The Environment Understanding layer emphasizes on understanding textual/semantic information from the visual inputs. Optical Character Recognition (OCR) is carried out with EasyOCR to extract the printed and hand written texts from documents, signboards, receipts and labels [17]. Preprocessing techniques such as denoising, contrast enhancement are applied to enhance the recognition accuracy as shown in Fig. 2. To further improve the situational awareness, the system uses BLIP (Bootstrapped Language Image Pretraining), a transformer-based vision-language model for scene understanding and image captioning. BLIP helps to interpret the visual details and object relationships to describe the surrounding environment in a meaningful way and represent users with a textual and contextual understanding of the surrounding environment. The Interaction and Communication layer is an interface between the system and the end user. All textual outputs that are created from the perception and environment understanding layers are processed into natural sounding speech using eSpeakNG, an offline efficient text-to-speech engine that was chosen for its low computational overhead and multilingual capabilities [18]. This is a guarantee of accessibility even in low connectivity environments. For conversational assistance, the system combines with Dialogflow which processes the user's voice commands using NLP techniques to identify intent and context [19]. Users can communicate with the system by requesting information, and the system responds via speech output and allows for natural and intuitive interaction. Facial images will be collected only with informed consent and stored securely. Initially, live video frames are taken and processed by the Perception and Recognition layer for obstacle detection, currency recognition and face and emotion analysis. The information that has been extracted is then passed on to the Environment Understanding layer for the text reading and scene interpretation.

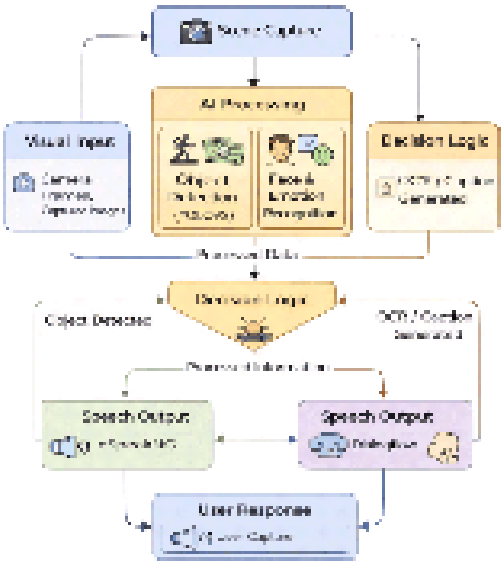


Fig.3 Flowchart of real-time operation

Finally, the Interaction and Communication layer is used to provide the processed information to the user with speech feedback and conversational interaction as shown in Fig. 3. This closed loop workflow combines perception, cognition, and communication in order to deliver responsive and real-time assistance.

I. Experimental Results and Discussion

The proposed Artificial Intelligence (AI) powered assistive system for visually impaired people were analysed to check the effectiveness and real-time performance of the integrated systems modules. Each component was then tested individually and then as part of the total system pipeline. The results illustrate that the modular integration does not cause any degradation in individual performance and allows smooth and low latency operation suitable for real-world assistive use [20].

Experimental evaluation was done on five important functionalities including obstacle and object detection, currency recognition, face and emotion recognition, text reading and scene captioning with speech output. The implementation of the system has been done in Python-Flask-OpenCV framework and tested in a Google Colab environment with a Tesla T4 GPU and VS Code environment. Some standard metrics like accuracy, recall, F1-score, and mean squared error (MSE) were employed to measure performance [21].

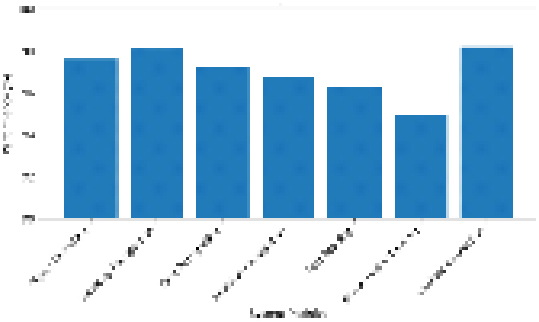


Fig.4 Performance Comparison of AI Modules

Table 1 Performance Of Individual Modules In The Proposed System

Module	Model	Evaluation Metrics	Accuracy %
Object and Obstacle Detection	YOLOv5	mAP@0.5	97.6
Currency Recognition	Darknet YOLOv4	Accuracy	98.1
Face Recognition	DeepFace	Accuracy	97.2
Emotion Recognition	DeepFace	F1-Score	96.8
Text Reading (OCR)	EasyOCR	CER	4.2
Scene Understanding	BLIP	F1-Score	94.9
Speech Output	eSpeakNG	Latency	200 ms
Chatbot Interaction	Dialogflow + NLP	Response Accuracy	98.2

The experimental results show that the proposed system works accurately and efficiently across all its modules as shown in Fig. 4. Object and obstacle detection achieved an accuracy of up to 97.6%, while currency recognition reached 98.1% even under varying conditions [22], [23]. Face and emotion recognition performed reliably with an F1-score of 96.8%, helping the system support social awareness [24]. Text recognition and scene understanding also handled real-world situations effectively [25], [26]. In addition, speech output using eSpeakNG showed low latency, and conversational interaction through Dialogflow remained accurate, enabling smooth user interaction. The system was tested in a controlled setting and future work will focus on mobile platforms.

The system was tested in ways. We checked how it worked in semi-outdoor environments. It worked well when the light was not too bright or too dim. The system was able to make decisions enough. However when it was very dark the system did not work well as it could. This means that the system might work better if we can make the pictures look better in light. The system had some problems when the light was very low so we think that making the pictures clearer in light might help the qualitative testing of the system and the quantitative evaluation of the system.

Figure 5 shows how the Indian currency recognition recognizes the notes under various conditions. Figure 6 shows the output of Scene captioning with a description of a women talking on a cell phone. Figure 7 shows the output of the identification and emotion recognition. These results show that the system gives the user fast and helpful in daily life.

TABLE 2 STATE OF THE ART COMPARISON OF CORE MODULES

Feature	Joshi et al.	Jaiswal et al.	Proposed Unified System
Indian Currency	Not Included	96.2% Accuracy	98.1% (2,050 Images)
Emotion Recognition	Not Included	Not Included	96.8% (F1-Score)
Scene Captioning	Navigation only	Not Included	BLIP Transformer Context
Speech Engine	Not Specified	Not Specified	eSpeakNG (200ms Latency)
Architecture	Single-Module	Single Module	Multi-module Unified

Fig.5 Output of Indian Currency Detection

```
Press 's' to capture image, 'q' to quit.
Image captured!
Caption: a woman talking on a cell phone
```

Fig.6 Output of Scene Captioning

```
Hello Ishwarya. You look neutral
```

Fig.7 Output of Identification and emotion recognition

I. CONCLUSION

This research work has dealt with the design, development, and analysis of an AI-assisted assistive tool intended to improve the autonomy, safety, and independence of daily activities of the visually impaired. The AI-assisted assistive tool combines computer vision, natural language processing, and speech technology into a single platform to address the major issues associated with autonomy and safety, such as text recognition, currency recognition, social awareness, and environmental understanding.

Contrary to the traditional assistive tool that normally requires independent assistance, the AI-assisted tool provides the visually impaired with assistive assistance in a timely and adaptive manner.The experiments show that the integrated system possesses high accuracy (94% –98%), low error rate performance simultaneously in real time.

By integrating YOLOv5-based object detection, Darknet YOLOv4-based currency recognition, face and emotion identification using DeepFace, EasyOCR for text, and BLIP for scene understanding, the system supports successful perception, understanding, and communication of the intelligent system in a single intelligent environment that encourages independent mobility and socialization. The system proposed here presents a trustworthy and scalable approach in the area of socially assistive AI and the possible benefits of combined AI solutions in facilitated living for persons who are visually impaired.

The future work is going to be about putting the system on things like smartphones and special watches so that people can carry it around easily. They also want to add feedback that you can feel and make the system understand lots of languages so that more people can use it. They will also try to make sure that the system is private and safe for people who use it especially when it comes to information like biometric data. The system will be tested with people who're visually impaired to see how well it really works and to make it better for the people who will be using it. Future work will focus on the system and it will be about the system and its portability, on embedded platforms and smartphones.

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PLAGIARISM REPORT

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Figure A2.3 PERCENTAGE OF AI DETECTED

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Figure A2.4 SIMILARITY REPORT

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Figure A2.5 OVERALL SIMILARITY

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